

Newton's Laws of Motion

Newton's First Law: an object remains at rest unless acted upon by a net external force. Second Law: $F = ma$. Third Law: every action has an equal and opposite reaction. Momentum $p = mv$ is conserved in closed systems. Kinetic energy $KE = 0.5mv^2$. The work-energy theorem relates work done to change in kinetic energy. Gravitational force $F = Gm_1m_2/r^2$.